

Tensor computations

Classification of perfect tensor formats with generic unique decomposition

Difficulty: extreme **Importance:** interesting to the community

Statement

Let $d \geq 3$, $n_1, \dots, n_d \geq 2$, and suppose the dimension-count value

$$r_* = \frac{\prod_{i=1}^d n_i}{1 + \sum_{i=1}^d (n_i - 1)}$$

is an integer. Such a format is called perfect. The conjecture asserts that a generic tensor in $\bigotimes_{i=1}^d \mathbb{C}^{n_i}$ has a unique minimal-length decomposition as a sum of rank-one tensors if and only if, after permuting factors, the format is

$$(2, k, k) \ (k \geq 2), \quad (3, 4, 5), \quad (2, 2, 2, 3).$$

Generic means on a nonempty Zariski-open subset of the entire tensor space. Rank one means $v_1 \otimes \dots \otimes v_d$ with nonzero complex factors. Two decompositions are the same when their rank-one tensor summands agree after permutation; reciprocal rescalings within a summand create no new decomposition. No symmetry constraints are imposed.

Relevance

This determines whether a decomposition algorithm can uniquely recover components of a generic tensor at the parameter-count threshold. The exceptional positive cases support explicit decomposition algorithms.

References

1. J. D. Hauenstein, L. Oeding, G. Ottaviani, and A. J. Sommese, *Homotopy techniques for tensor decomposition and perfect identifiability*, J. reine angew. Math. 753 (2019), 1–22. DOI; [primary preprint](#), §1, equation (2) and Conjecture 1.3, p.3; Theorems 1.1–1.2 establish the two sporadic positive cases.
2. A. Massarenti, M. Mella, and G. Staglianò, *Effective identifiability criteria for tensors and polynomials*, J. Symbolic Comput. 87 (2018), 227–237. [Author PDF](#), §1, discussion of the Hauenstein et al. generic-identifiability conjecture.

Status check — 2026-09-08

The latest originating preprint revision (November 9, 2016; journal publication 2019) retains Conjecture 1.3, while replacing its former symmetric analogue by Galuppi–Mella’s theorem. Searches “*perfect identifiability*” *conjecture* 2025 2026, “*perfect formats*” “*conjecture*” “2025” OR “2026”, and “*Hauenstein*” “*identifiability*” “*conjecture*” “2026” located no resolution of this nonsymmetric statement. This is a bounded search without a new 2026 reaffirmation.